Recursive Self-Modeling Cognition: An Objective Framework for Identity, Memory, and Perception

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Abstract

We present a formal cognitive architecture that models thought and identity as nested, self-referential feedback loops. The framework unifies (i) a Perceptual Loop that encodes signals across time, (ii) an Integrative Loop that weighs and links representations with memory, and (iii) an Identity Loop that updates a self-state through a generative function driven by prediction error. We formalize the update equations, specify a tagging-and-recall memory mechanism, and provide falsifiable predictions and evaluation protocols. The approach aligns with cybernetics, control theory, and predictive processing while emphasizing "error as growth." We argue the model is suitable both for human data (journaling, audio) and artificial agents (self-modeling AI), and we outline limitations and ethical considerations.

1 Introduction

Identity is not a static object; it is a process. This report formalizes a recursive account of identity formation in which a system iteratively models itself, updates expectations, and integrates new evidence. Two complementary artifacts motivated this work: (a) a mathematical Overleaf draft describing nested loops (Perceptual → Integrative → Identity), and (b) handwritten notes emphasizing that "every core idea expands outward but is constantly monitored, tagged, and recalled." We fuse these perspectives into a single architecture—the *Recursive Self-Modeling (RSM)* theory.

2 Background and Related Work

RSM intersects several literatures: cybernetics and feedback control; predictive coding and active inference; hierarchical working memory; and Hofstadter's notion of "strange loops." RSM inherits the feedback discipline of control, the emphasis on prediction errors from predictive processing, and the reflexive, self-descriptive hallmark of strange loops. Unlike purely statistical accounts, RSM adds an explicit, inspectable memory tagging and recall substrate intended for human-interpretable auditing.

3 Notation and Preliminaries

Time is discrete: $t = 0, 1, 2, \ldots$ Let $S(t)$ denote the self-state at time t. Incoming signals $x(t)$ are encoded by the Perceptual Loop into temporal integrators $a_t(t)$. The Integrative Loop produces an emergent identity contribution $a_e(t)$ using weighted branches $B_i(t)$, links $L_i(t)$, and retrieved memory $M_r(t)$. Prediction error $\varepsilon(t)$ quantifies mismatch between expected and observed signals. The Identity Loop updates $S(t+1)$ using a generative function $\mathcal{G}$.

4 Model

4.1 Perceptual Loop

The Perceptual Loop encodes streams $x(t)$ into temporally smoothed features $a_t(t) = U(t)$. U may be a moving average, exponential filter, or learned encoder. Output $a_t(t)$ feeds the Integrative Loop.

4.2 Integrative Loop

The Integrative Loop weights and links current features with recalled memory. Let $w(t)$ be attention weights over branches $B_i(t)$ and links $L_i(t)$. We write

$$a_e(t) = C \cdot w(t)^\top y(t) - \lambda D(t), \tag{1}$$

where $y(t)$ concatenates candidate branches/links, C is a scaling constant, and $D(t)$ is a dispersion term. Memory returns $M_r(t)$ by tag-based lookup.

4.3 Identity Loop

The core update is:

$$S(t+1) = \mathcal{G}\Big(a_t(t) + a_e(t+1)\Big). \tag{2}$$

Here $a_e(t+1)$ is the integrative output after incorporating prediction error $\varepsilon(t)$.

4.4 Error as Growth

Prediction error is not merely noise. Let $\varepsilon(t) = x(t) - \hat{x}(t)$, where $\hat{x}(t)$ is the system's expectation. We treat $\varepsilon(t)$ as a growth signal that biases $w(t+1)$ toward under-represented branches and tags new memory with the experienced discrepancy.

4.5 Memory Tagging and Recall

Each node (idea) stores a tuple $\langle \text{content}, \text{tags}, \text{timestamp}, \varepsilon\text{-trace} \rangle$. Tags include semantic labels, loop-of-origin, and task context. Recall is content-addressable by tags, so new inputs retrieve prior nodes with shared tags; retrieved content refines $a_e(t)$ and thus $S(t+1)$.

4.6 State-Space View

Let $X(t) = [S(t), a_t(t), a_e(t), w(t), M_r(t)]^\top$. Then the RSM dynamics can be written compactly as:

$$X(t+1) = F(X(t), x(t), \varepsilon(t)), \tag{3}$$

with output $S(t+1)$.

5 Algorithms

5.1 Identity Update (per step)

```
INPUT: x(t), memory M, previous state S(t)
1: a_t <- U(x(t))
2: T <- tagset(x(t), S(t))                  # derive tags
3: M_r <- recall(M, T)                      # content-addressable retrieval
4: y <- concat(B(t), L(t), M_r, a_t)        # candidate context
5: eps <- predict_error(x(t), x_hat(t))     # deviation signal
6: w <- attn(y, eps)                        # attention weights modulated by
   error
7: a_e <- C * dot(w, y) - lambda * D(y)     # integrative output
8: S_next <- G( a_t + a_e )                 # identity update
9: write(M, S_next, tags=union(T, {eps}), time=t)
RETURN S_next, M
```

6 Theoretical Properties

- **Directed Plasticity:** Increasing ε on a class of inputs increases future attention to branches linked to that class.
- **Stability–Plasticity Trade-off:** λ controls dispersion $D(y)$; high $\lambda \Rightarrow$ stable identity, low $\lambda \Rightarrow$ rapid adaptation.
- **Strange-Loop Self-Consistency:** Under bounded ε and stationary tags, $S(t)$ approaches a self-consistent attractor.

7 Evaluation Protocols

1. **Prediction Tasks:** forecast next-step sensory signals; metric = NRMSE and ε-profile.
2. **Identity Stability:** perturbation tests on autobiographical prompts; metric = self-consistency score.
3. **Recall Utility:** ablate memory recall; metric = change in task performance and identity drift.

8 Applications

- Self-modeling AI agents with interpretable identity states.
- Therapeutic journaling analytics that visualize ε-driven growth.
- Robotics where identity encodes mission role and safety constraints.
- Augmented cognition for co-adaptive human–AI systems.

9 Limitations and Risks

RSM is not a full theory of consciousness. Identity attractors could become rigid (over-regularized λ) or self-confirming (biased tags). Memory privacy is paramount; audit trails must be secured. Empirical validation is required.

10 Conclusion

RSM offers a compact, testable account of identity as a nested, recursive computation in which prediction error fuels growth and memory supplies structure. The model is implementable, auditable, and compatible with control-theoretic tools.

A Equation Summary

$$
\begin{aligned}
a_t(t) &= U(x(t)) \\
a_e(t) &= C \cdot w(t)^\top y(t) - \lambda D(t) \\
S(t+1) &= \mathcal{G}(a_t(t) + a_e(t+1)) \\
\varepsilon(t) &= x(t) - \hat{x}(t) \\
X(t+1) &= F(X(t), x(t), \varepsilon(t))
\end{aligned}
$$

B Glossary

Symbol	Definition
$S(t)$	Self-state at time t
$a_t(t)$	Temporal integrator from perceptual loop
$a_e(t)$	Emergent identity contribution from integrative loop
$\varepsilon(t)$	Prediction error; driver of plasticity
$w(t)$	Attention weights
$D(t)$	Dispersion regularizer
$\mathcal{G}(\cdot)$	Generative update function

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